

5.3 Transition elements

This section provides learners with a deeper knowledge and understanding of the periodic table within the context of the transition elements.

This section includes the role of ligands in complex ions, stereochemistry, precipitation, ligand substitution and redox reactions. The colour changes and observations in these reactions increase the toolkit of qualitative inorganic tests for identifying unknown ionic compounds.

5.3.1 Transition elements

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Properties	
(a) the electron configuration of atoms and ions of the d-block elements of Period 4 (Sc–Zn), given the atomic number and charge (see also 2.2.1 d)	Learners should use sub-shell notation e.g. for Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$.
(b) the elements Ti–Cu as transition elements i.e. d-block elements that have an ion with an incomplete d-sub-shell	
(c) illustration, using at least two transition elements, of: (i) the existence of more than one oxidation state for each element in its compounds (see also 5.3.1 k) (ii) the formation of coloured ions (see also 5.3.1 h, j–k) (iii) the catalytic behaviour of the elements and their compounds and their importance in the manufacture of chemicals by industry (see 3.2.2 d)	No detail of how colour arises required. Practical examples of catalytic behaviour include: Cu ²⁺ for reaction of Zn with acids; MnO ₂ for decomposition of H ₂ O ₂ . No detail of catalytic processes required. HSW9 Benefits of reduced energy usage; risks from toxicity of many transition metals.

Ligands and complex ions

- (d) explanation and use of the term *ligand* in terms of coordinate (dative covalent) bonding to a metal ion or metal, including bidentate ligands
- (e) use of the terms *complex ion* and *coordination number* and examples of complexes with:
- (i) six-fold coordination with an octahedral shape
 - (ii) four-fold coordination with either a planar or tetrahedral shape (**see also 2.2.2 g–h**)
- (f) types of stereoisomerism shown by complexes, including those associated with bidentate and multidentate ligands:
- (i) *cis–trans* isomerism e.g. $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ (**see also 4.1.3 c–d**)
 - (ii) optical isomerism e.g. $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ (**see also 6.2.2 c**)
- (g) use of *cis-platin* as an anti-cancer drug and its action by binding to DNA preventing cell division

Examples should include:
monodentate: H_2O , Cl^- and NH_3
bidentate: $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ ('en').

In exams, other ligands could be introduced.

M4.1, M4.2

Examples:
Octahedral: many hexaaqua complexes, e.g. $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
Tetrahedral: many tetrachloro complexes, e.g. CuCl_4^{2-} and CoCl_4^{2-}
Square planar: complexes of Pt, e.g. *platin*: $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$ (**see also 5.3.1 g**).

M4.1, M4.2, M4.3

Learners should be able to draw 3-D diagrams to illustrate stereoisomerism.

HSW8

HSW9 Benefits of chemotherapy; risks from unpleasant side effects.

Ligand substitution

- (h) ligand substitution reactions and the accompanying colour changes in the formation of:
- (i) $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ and $[\text{CuCl}_4]^{2-}$ from $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
 - (ii) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ from $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (**see also 5.3.1 j**)
- (i) explanation of the biochemical importance of iron in haemoglobin, including ligand substitution involving O_2 and CO

Complexed formulae should be used in ligand substitution equations.

Precipitation reactions

- (j) reactions, including ionic equations, and the accompanying colour changes of aqueous Cu^{2+} , Fe^{2+} , Fe^{3+} , Mn^{2+} and Cr^{3+} with aqueous sodium hydroxide and aqueous ammonia, including:
- (i) precipitation reactions
 - (ii) complex formation with excess aqueous sodium hydroxide and aqueous ammonia

For precipitation, non-complexed formulae or complexed formulae, are acceptable e.g. $\text{Cu}^{2+}(\text{aq})$ or $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$; $\text{Cu}(\text{OH})_2(\text{s})$ or $\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4$.
With excess NaOH, only $\text{Cr}(\text{OH})_3$ reacts further forming $[\text{Cr}(\text{OH})_6]^{3-}$.
With excess NH_3 , only $\text{Cr}(\text{OH})_3$ and $\text{Cu}(\text{OH})_2$ react forming $[\text{Cr}(\text{NH}_3)_6]^{3+}$ and $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ respectively (**see also 5.3.1 h**).

Redox reactions

- (k) redox reactions and accompanying colour changes for:
- (i) interconversions between Fe^{2+} and Fe^{3+}
 - (ii) interconversions between Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$
 - (iii) reduction of Cu^{2+} to Cu^+ and disproportionation of Cu^+ to Cu^{2+} and Cu
- (l) interpretation and prediction of unfamiliar reactions including ligand substitution, precipitation, redox.

Fe^{2+} can be oxidised with $\text{H}^+/\text{MnO}_4^-$ and Fe^{3+} reduced with I^- , Cr^{3+} can be oxidised with $\text{H}_2\text{O}_2/\text{OH}^-$ and $\text{Cr}_2\text{O}_7^{2-}$ reduced with Zn/H^+ , Cu^{2+} can be reduced with I^- . In aqueous conditions, Cu^+ readily disproportionates.

Learners will **not** be required to recall equations but may be required to construct and interpret redox equations using relevant half-equations and oxidation numbers (**see 5.2.3 b–c**).